

Project Proposal: EduGraph

Group 6:

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A. Assigned Paradigm & Tool

The paradigm assigned for this project is the Graph Database. Graph databases prioritize the relationships between data points just as much as the data itself, using nodes and edges to represent complex, highly interconnected networks. For this project, we will be utilizing Neo4j (and/or Memgraph) as our primary database engine to build and query our knowledge graph.

B. Real-World Case Study

Case Study: NASA's Knowledge Graph Implementation

Objective & Problem

NASA faced a critical "data silo" challenge where decades of mission reports, technical expertise, and "lessons learned" were fragmented across disconnected relational databases. This fragmentation led to engineers spending nearly 30% of their time searching for information and risked the loss of vital "tribal knowledge" as veteran staff retired.

Solution & Implementation

NASA adopted a Graph Database paradigm (using Neo4j and Memgraph) to build a unified Knowledge Graph. Unlike traditional databases, this model treats relationships as first-class citizens, allowing NASA to map complex connections between engineers, mission phases, technical parts, and historical documents. This architecture utilizes Natural Language Processing (NLP) to convert unstructured PDF reports into a searchable, interconnected network.

Key Findings

- **Accelerated Decision-Making:** The graph system significantly reduced information retrieval time, allowing for faster, data-driven safety checks for upcoming missions like the Artemis program.
- **Expertise Discovery:** By applying graph algorithms (such as Community Detection), NASA can identify "hidden" experts based on their involvement in past projects and documentation, rather than just official job titles.
- **Proactive Risk Mitigation:** Connecting "Lessons Learned" directly to specific mission phases ensures that past engineering failures are visible and avoidable for

current teams, effectively bridging the gap between historical data and future exploration.

C. Creative Project Idea

Title : EduGraph | Academic Knowledge & Career Path Navigator

Domain : Higher Education and Career Navigation

D. Problem Statement

EduGraph is a graph-based academic navigation system designed to help university students discover meaningful connections between their coursework, technical skills, and career trajectories.

- **Fragmented Academic Information.** Course curricula, prerequisite structures, faculty profiles, and alumni records exist in isolated systems with no interconnection, making holistic academic planning difficult.
- **Invisible Prerequisite Dependencies.** Course prerequisite chains are rarely visualized in a way that reflects their true depth and complexity. Students frequently encounter academic setbacks caused by incomplete understanding of how foundational courses relate to advanced ones, leading to inefficient course sequencing decisions.
- **Disconnected Skill to Career Mapping.** There is no structured mechanism that maps the technical skills developed through specific courses to the competencies demanded by industry roles. Students are left to independently infer which courses are most professionally relevant, often without access to reliable empirical data.
- **Inaccessible Alumni Insights.** Alumni career trajectories remain an untapped resource, as no system captures and exposes their validated academic-to-career paths for current students to query.
- **Underutilized Faculty-Research Connections.** Faculty research interests are not linked to course content or student goals, preventing effective identification of potential academic mentors.

University students face a problem identical to NASA. Information is fragmented across different systems. Curriculum data is in one place while prerequisite maps are often invisible. Career data and alumni profiles are on LinkedIn and job portals. Students cannot easily answer complex questions like which courses are most crucial to becoming a Machine Learning Engineer or which alumni from their faculty are already on that career path. EduGraph connects these dots into a single navigable map for students and faculty.

E. Conceptual Data Model

The structure uses nodes to represent academic and professional entities and edges to represent their dependencies.

Node Types

Node Label	Key Properties
Course	code, name, credits, semester
Skill	name, category
Career	position, industry
Alumni	name, graduation_year, company
Faculty	name, research_interest
Company	name, sector

Relationship Types (Edges)

Relationship	Meaning
PREREQUISITE_OF	Course dependency chain
BUILDS_SKILL	Skill acquired through course
REQUIRED_FOR	Skill required for a career role
COMPLETED	Course taken by alumni
WORKS_AS	Alumni's current/past role
WORKS_AT	Alumni's employer
TEACHES	Faculty teaching assignment
RESEARCHES	Faculty research focus

F. Role Distribution

Name	Role
Soraya Azzizah Pahlevi	Frontend Developer & Graph DB Architect
Haitsam Ahmad Fakhri	Frontend Developer & Database Integrator
Fahmi Milan Amyar	Backend Developer & System tester

Alicia	Backend Developer & System tester
Rain Elgratio	Backend Developer & System tester